

Effect of 3D printed replicas on the duration of third molar autotransplantation surgery: A controlled clinical trial

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Abstract

Background/Aim: This study aimed to develop a protocol that combines cone-beam computed tomography (CBCT), software, and 3D printing to design 3D replicas for tooth autotransplantation. The goal was to evaluate the impact of this approach on the extraoral time of the donor teeth and the total surgical time, thereby enhancing surgical efficiency and outcomes.

Materials and Methods: A non-randomized trial (protocol [10.1186/ISRCTN13563091](https://doi.org/10.1186/ISRCTN13563091)) was conducted at Riga Stradins University, enrolling 46 patients (13–22 years old) who required molar extraction and possessed a non-erupted third molar. The patients were sequentially assigned to a 3D replica group (24 patients) or a control group (22 patients). The primary outcome measured was the extra-alveolar time of the donor tooth, and the secondary outcome was the total duration of surgery. Both were assessed using a sample size capable of detecting a 10-min difference. A generalized linear model adjusted for various factors was used to test for significant time differences ($p < .05$) between the groups.

Results: Forty-six patients were included in this analysis. The effect of using 3D replicas was not statistically significant and was associated with a decrease in the extraoral time of the donor tooth in seconds ($\beta = -9.35$, 95% CI [-40.86, 22.16]). For the total surgical time in minutes, the use of 3D replicas had a statistically significant impact, reducing the operation duration in minutes ($\beta = -13.42$, 95% CI [-24.50, -2.34]). No early complications were observed in either group, with all teeth present at 3–4 weeks post-surgery.

Conclusions: The integration of 3D printing technology can enhance the efficiency of autotransplantation surgeries, primarily by reducing surgical time.

KEYWORDS

cone-beam computed tomography, immature, printed tooth replica, third molar, tooth transplantation

1 | INTRODUCTION

Autotransplantation, or autogenous tooth transplantation, involves moving a donor tooth to a recipient site. Tooth autotransplantation is a successful alternative to dental implants for tooth replacement due to dental caries or trauma, with success rates of over 95% in children and adolescents, and 80% in adults.¹ Intentional tooth replantation also has a high success rate of 89.1% and offers a more cost-effective alternative, with the option for later implant placement if needed.² This method is especially beneficial in growing children, where implants are typically not recommended.^{3,4} Additionally, autotransplantation offers advantages such as avoiding the preparation of adjacent teeth, comparative cost-effectiveness, and arguably serves as a better alternative to fixed or removable prostheses.⁵ The indications for autotransplantation include premature or traumatic tooth loss, impacted, ectopic or missing teeth, tumors, iatrogenic injury, and poor prognosis.⁵ Autotransplantation involves extracting the tooth, preparing the recipient site, atraumatic extraction of the donor tooth, minimizing the extra-alveolar time of the donor tooth, positioning and stabilizing the donor tooth, and making occlusal adjustments.⁶ Aside from extraoral time, several other factors critically influence the viability of autotransplantation.⁷ These include potential damage to the donor tooth during extraction, the need for multiple fitting attempts at the recipient site, and the preparation quality of the recipient site. All these factors can substantially affect the viability of the periodontal ligaments.⁸

Current research suggests that the 5-year survival rate of autotransplantation is between 81% and 98.2%,⁹ but evidence from controlled clinical studies remains limited.⁹ Autotransplantation of teeth with incomplete root formation further supports this, showing survival rates of 97.4% after 1 year, 97.8% after 5 years, and 96.3% after 10 years, making it a reliable treatment option with low risks.¹⁰ The survival rate of a donor tooth transplant is influenced by its extraoral time.^{11,12} Shorter durations enhance the viability of periodontal ligament cells.^{12,13} However, extraoral time is not the sole determinant. The quality of donor tooth extraction, number of fitting attempts, and recipient site preparation are equally crucial for maintaining periodontal ligament viability.¹⁴ To this end, 3D-printed replicas reduce trauma,¹⁵ extraoral exposure, and potentially preserve cell viability.^{10,16} Moreover, 3D-printed replicas have shown promise in mitigating some of these factors. For example, they can help minimize damage to the donor tooth during extraction and reduce the number of fitting attempts required, thus preserving the integrity of the periodontal ligament.¹⁷⁻¹⁹ Case reports have documented the use of cone-beam computed tomography (CBCT)-created artificial sockets and 3D-printed donor tooth replicas.¹⁶ Methodologies include using CBCT-created artificial sockets,¹⁷ surgical guides,^{20,21} low-cost desktop 3D printed resin replicas,^{22,23} and stainless steel templates.²⁴

Despite promising results such as a reported average reduction in operative time of 7.7 min,¹⁸ most studies have not included a control group.⁶⁻⁸ Thus, much of the existing evidence relies on case reports without objective comparisons. The only clinical study

currently available suggests that a CBCT-guided approach improves the predictability of autotransplantation.¹⁷

Clinical evidence is sparse, and there is a lack of established guidelines for the effective use of CBCT, 3D printing, and computer-assisted design techniques. Pre-planning autotransplantation with 3D printing has extensive advantages, as it is a safe, biocompatible, and reliable procedure with high survival rates and satisfactory aesthetics.²³ However, the cost and resource-intensive nature of replica production, coupled with the additional radiation dose received by the patient during CBCT examination, necessitates objective evaluations for efficiency in real surgical settings. We aimed to develop a comprehensive protocol combining these technologies to design 3D replicas for autotransplantation and conduct a clinical trial to answer the following research questions: (1) Does the use of 3D-printed replicas in dental autotransplantations decrease extraoral and total surgical times? (2) Does the use of these replicas improve surgical efficiency? We hypothesized that 3D-printed replicas in dental autotransplantations would significantly decrease the extraoral and total surgical times, thereby improving surgical efficiency.

2 | MATERIALS AND METHODS

2.1 | Trial design, report, and implementation

A non-randomized controlled clinical trial was conducted. This study adhered to the principles of the Declaration of Helsinki. A detailed protocol was developed and submitted to the Ethics Committee of Riga Stradins University, which approved the study (approval Nr. 6-1/08/12). The protocol has been published and is available at ISRCTN13563091 <https://doi.org/10.1186/ISRCTN1356309>.²⁵ The study was implemented according to the published protocol and reported following the CONSORT recommendations (Table S1).²⁶

2.2 | Participants

From August 2019 to September 2022, 46 patients aged 13-22, requiring permanent premolar, first, or second molar extraction, and having at least one non-erupted third molar were referred to Riga Stradins University's Institute of Stomatology. Each invited participant received a thorough briefing on the trial objectives, procedures, potential risks, and benefits. This ensured that the informed consent was comprehensively understood and provided in writing. They were also reassured of their right to withdraw at any point without negative consequences. Enrollment in the study required participants to meet the following inclusion criteria: (1) age < 25 years; (2) missing at least one premolar or molar, planned to be replaced by an autotransplanted third molar; (3) comprehension of the procedure, along with provision of written informed consent; (4) non-erupted third molars irrespective of root anatomy (Moorrees development

stages 3–5,²⁷ Figure S1); (5) clinically healthy and not pregnant; (6) good oral hygiene (no gingivitis); and (7) non-smoking status. All 46 invitees met the inclusion criteria and were enrolled in the study. The exclusion criteria included a history of traumatic extraction of the donor tooth. Patients who could not attend the scheduled control visits were excluded from the study.

2.3 | Allocation of participants

The patients were sequentially assigned to groups based on the limited availability of CBCT, with the replica group prioritized during the designated CBCT slots.

2.4 | Interventions

Patients assigned to the 3D group had a maxillary CBCT radiographic examination (i-CAT Next Generation, Kavo, Germany) with a voxel size of 0.3 mm, tube voltage of 120 kV, current of 5 mA, and an exposure time of 4 s using a field of view of 13 × 16 cm. A 3D-printed replica of the third molar, the tooth to be transplanted, was created according to the protocol available in the Data S1. Patients in the control group did not undergo a CBCT examination.

Recipient-site analysis was performed for each patient. All patients had sufficient vertical height to the mandibular nerve. Lateral defects were present, but could be addressed by splitting the bone. Many patients had insufficient vertical height of the upper jaw. When the bone deficit relative to the maxillary cavity ranged from 1 to 3 mm, we performed an osteotomy using Xive osteotomes (implant tapping technique). For deficits of 4 mm or more, a sinus lift procedure was performed, with the osteotomy window positioned occlusally.

One surgeon with experience in performing over 20 autotransplantations prior to this study planned and executed all the procedures. Due to the distinct methods involved, blinding the surgeon to the intervention type was unfeasible. All patients received amoxicillin/clavulanic acid 875 mg/125 mg x2 (maximum 20 mg/kg body weight) or 600 mg clindamycin (maximum 15 mg/kg body weight) preoperatively in cases of a prior allergy to penicillin. No additional antibiotics were administered. In the recipient region, osteotomy was performed using a straight fissure and surgical round burs with implant burs from Straumann BLT (Basel, Switzerland). Cooling was maintained using saline from an implanted Kavo Physio dispenser (KaVo, Biberach an der Riss, Germany), and the drill speed was limited to 800 rpm. Once the extracted tooth socket was adapted, the autotransplanted tooth was extracted and processed as usual.⁴ All donor teeth were immature, unerupted, Moorrees stages (3–5) Root ½ till open apex. The donor tooth was extracted and promptly inserted into the prepared osteotomy or alveolus, being positioned in supragingival infraocclusion. Suturing was performed with either Silk 3-0 or Vicryl 4-0, depending on the surgical requirements. The 3D tooth

replica model was adjusted to the recipient site prior to autotransplantation. In the control group, in which a 3D-printed replica was not used, the extracted tooth was re-inserted into the original alveolus for comparison. Patients in the control group underwent autotransplantation without a 3D replica. The extracted tooth was placed back into the socket if it did not fit well with the alveolus. When the donor tooth was located in the alveolus, the recording of the extra-alveolar time was paused. All the steps and 3D replicas are shown in Figure 1.

2.5 | Outcomes

The primary outcome was the difference in extra-alveolar socket time (in seconds) between the control and 3D-printed replica groups. Surgical time, measured in minutes from the first incision to the last suture, was the secondary outcome. The nurses responsible for recording the extra-alveolar and total surgical times were blinded to the group assignments to avoid detection bias. They were not involved in the surgical procedures and only received the time data for logging. At the same time, the surgeon was responsible for recording any complications such as infection, loss of the transplanted tooth, or other adverse effects.

2.6 | Variables recorded

Periapical radiography was performed on all patients after autotransplantation. Based on these images, the surgeon (ML) and an experienced radiologist (LN) assessed the tooth development stage based on mutual agreement using Moorrees' development stages.²⁷ Demographic data for each patient were also recorded. These variables were used for statistical adjustments to isolate the specific effects of the 3D replication. Another variable was the presence of a tooth at the recipient site.

2.7 | Sample size

The sample size for this study was calculated as 42 patients, divided into two groups of at least 21 patients each. This sample size was sufficient to detect a difference of 10 min in the total surgery time between the two groups, assuming a 5% alpha level, a statistical power of 80%, and an anticipated dropout rate of 10%. The sample size calculation was performed using the following formula:
$$n = \frac{\left(\frac{Z_{1-\alpha}}{2} * s_1 + Z_{1-\beta} * s_2\right)^2 * (s_1^2 + s_2^2)}{(m_1 - m_2)^2}$$
, where n is the sample size, $\frac{Z_{1-\alpha}}{2}$ is the $\frac{1-\alpha}{2}$ quantile of the standard normal distribution, $Z_{1-\beta}$ is the $1-\beta$ quantile of the standard normal distribution, s_1 is the standard deviation of the control group, s_2 is the standard deviation of the treatment group, m_1 is the mean of the control group, and m_2 is the mean of the treatment group.²⁸ The time of 10 min corresponds to the average reported in the available case reports.¹⁶

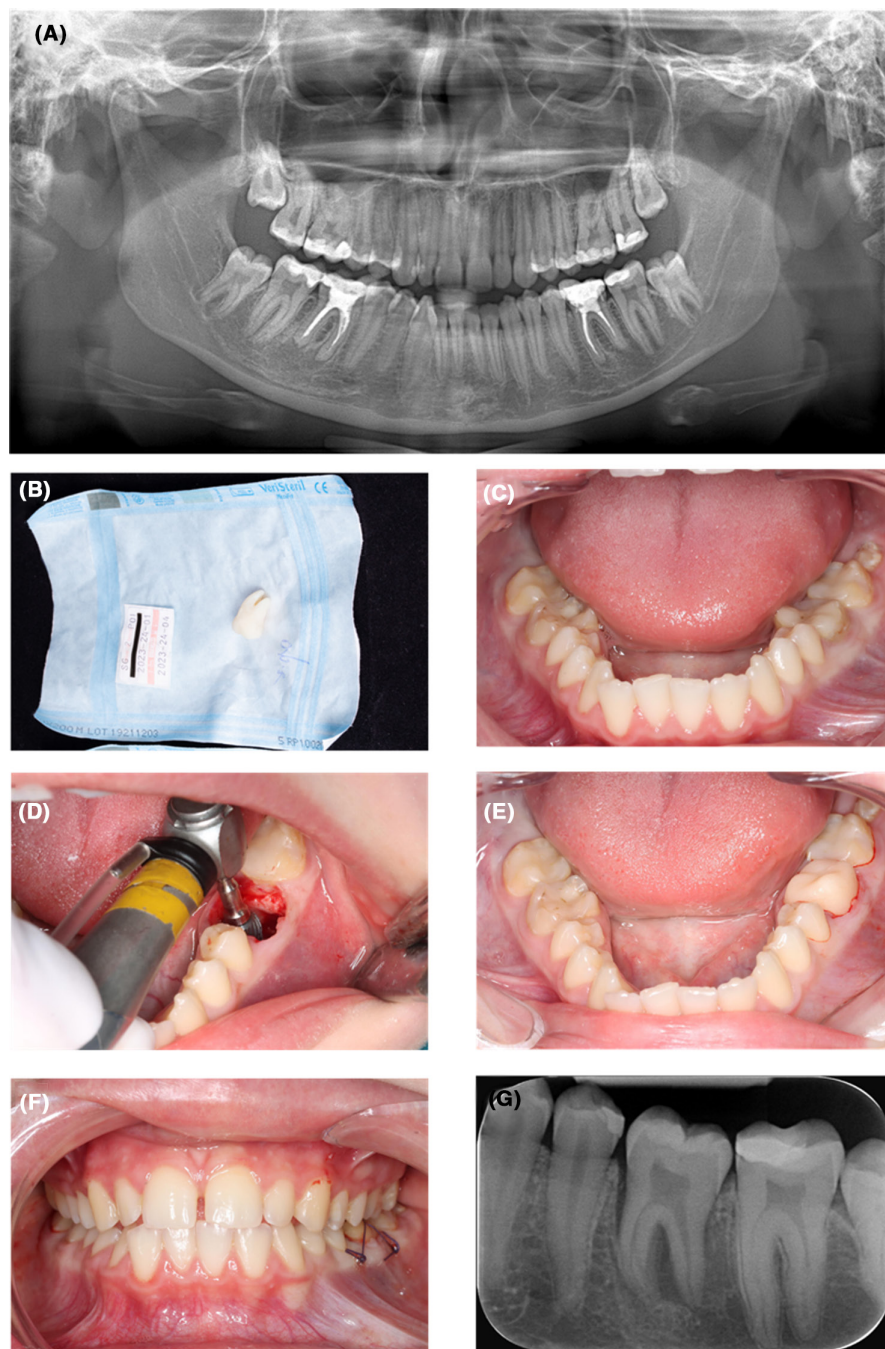


FIGURE 1 Surgical procedure. Autotransplantation of an immature third molar d48 to the d36 site. (A) Orthopantomogram of a 17-year-old female before treatment. The lower left first molar had a poor prognosis. The lower left first molar was replaced with an unerupted lower right third molar. The donor tooth had root development at stage 5 (Moorrees 1963), with the apex open by 1 mm. (B) The 3D-printed model was created according to the protocol (available in the Data S1). The replica was sterilized at 121°C in two-atmosphere steam sterilization (60 min sterilization cycle). (C) Occlusal view before autotransplantation. (D) After extraction, saline-irrigated burs were used to prepare the recipient site. (E) Try-in of the donor tooth replica. (F) The transplant is fixed using sutures. (G) Periapical radiograph obtained on the day of the surgery.

2.8 | Statistical methods

Descriptive statistical analyses were performed. The differences in the extraoral alveolar time in seconds and total surgical time in minutes between the groups were evaluated using a generalized linear model, adjusting for sex, age, quadrant of the transplanted tooth, and stage of development of the donor tooth root. Standardized parameters were obtained by fitting the model to a standardized version of the dataset. Confidence intervals (95% CIs) and *p*-values were computed using the Wald *t*-distribution approximation. The significance level was set at 5%. All analyses were performed using R statistical software (version 4.2.0).²⁹

3 | RESULTS

3.1 | Study population

Eighty-four patients were screened between August 2019 and September 2022, 58 of whom met the eligibility criteria. Twenty-eight patients were assigned to the control group and 30 to the 3D replica group, and 46 patients completed the trial. Figure 2 shows the CONSORT flow diagram that visually represents the study participants' enrollment, allocation, follow-up, and data analysis.

The 3D replica group comprised 24 patients (Figure 1). There were 16 (73%) women in the control group and 13 (54%) in the 3D replica

FIGURE 2 Consort flow diagram.

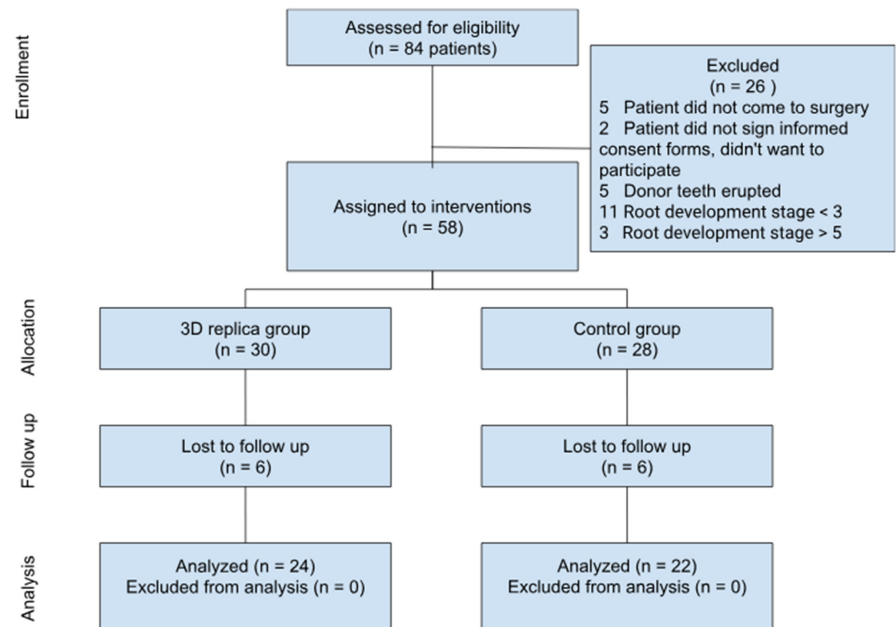


TABLE 1 Baseline characteristics of study participants.

Characteristic	Control, N = 22 ^a	3D replica, N = 24 ^a
Gender		
Female	16 (73%)	13 (54%)
Male	6 (27%)	11 (46%)
Age at surgery	17.82 (2.08)	16.46 (1.25)
Moorrees' development stages		
3	8 (36%)	9 (38%)
4	6 (27%)	12 (50%)
5	8 (36%)	3 (12%)
Donor > recipient site		
Mandibular > mandibular	10 (45%)	10 (42%)
Mandibular > maxilla	1 (4.5%)	0 (0%)
Maxilla > mandibular	8 (36%)	7 (29%)
Maxilla > maxilla	3 (14%)	7 (29%)

^an (%); Mean (SD).

group. The mean age (standard deviation [SD]) at the time of surgery in the control group was 17.8 (2.08) years, and that of the 3D replica group was 16.46 (1.25) years. The most common donor-recipient combination was mandibular-mandibular (control, 45%) (replicas, 42%) (Table S1). We observed no sign of significant changes in tooth size between the CBCT scan and surgery, which took up to 3 months. All replicas fit accurately during surgery. The demographic data and distribution of Moorrees' developmental stages are shown in Table 1.

3.2 | Surgical time outcomes

The mean (SD) extraoral time for the donor tooth in the control group was 67.7 (51.8) s, while in the 3D replica group, it was 52.7 (40.4) s.

The number of fitting attempts did not differ significantly between the groups (Table S2). The time difference was compared using a generalized linear regression model, adjusting for sex, age at surgery, developmental stage, and donor-recipient combination. In evaluating the extraoral donor time, the model revealed a nonsignificant reduction of about 9.35 s ($\beta = -9.35$, 95% CI [-40.86, 22.16], $p = .561$) when using a 3D replica, accounting for 19% of the observed variance ($R^2 = .19$). For total surgery time, the use of the 3D replica led to a statistically significant decrease of 13.42 min ($\beta = -13.42$, 95% CI [-24.50, -2.34], $p = .018$), with the model explaining 40% of the variance ($R^2 = .40$). Details of the fitted regression models are presented in Table 2. No complications or adverse effects were observed in either group. All teeth were present in the mouth 3–4 weeks after surgery.

4 | DISCUSSION

Autotransplantation is typically indicated for congenitally missing teeth, non-maintainable traumatized teeth, teeth with a compromised prognosis due to periodontal, restorative, or endodontic reasons, and developmental abnormalities.¹ In countries with a high prevalence of caries,³⁰ autotransplantation presents a viable option for rehabilitating patients with extensive damage. For instance, the prevalence of caries among 12-year-olds in Latvia has reached 71.9%, encompassing dental cavitated lesions, restored teeth, and those lost due to caries.³¹ While dental implants are commonly used, other treatment options, such as autotransplantation, should be considered for compromised or diseased teeth, as they offer competing success rates and longer-term retention in the dentition.³ Our study aimed to evaluate the impact of a comprehensive CBCT, software, and 3D-printing protocol on extraoral and total surgical time in autotransplantation. Although we did not find a significant reduction in the extraoral time of the donor

TABLE 2 Adjusted regression model for extraoral donor time (in seconds) and total surgery time (in minutes).

Characteristic	Extraoral donor time (s)			Total surgery time (m)		
	β	95% CI	p-Value	β	95% CI	p-Value
Group						
Control	—	—		—	—	
3D replica	-12	-42, 17	.4	-16	-26, -5.2	.006
Age at surgery	2.6	-6.7, 12	.6	2.0	-1.3, 5.3	.2
Sex						
Female	—	—		—	—	
Male	6.6	-23, 36	.7	2.8	-7.5, 13	.6
Moorrees' development stages						
3	—	—		—	—	
4	-7.4	-39, 24	.6	4.6	-6.6, 16	.4
5	-13	-49, 22	.5	5.2	-7.6, 18	.4
Donor > recipient site						
Mandibular > mandibular	—	—		—	—	
Mandibular > maxilla	121	24, 217	.019	48	13, 82	.010
Maxilla > mandibular	1.0	-30, 32	>.9	-90	-12, 10	.9
Maxilla > maxilla	.65	-35, 37	>.9	1.6	-11, 14	.8

Abbreviation: CI, confidence interval.

tooth, there was a significant decrease in the total surgical time, suggesting improved surgical efficiency, which can be attributed to the use of the 3D-printed replicas. The time saved during the fitting of the donor tooth contributed to a reduced overall extraoral time in the recipient socket for the study group, further supporting this conclusion. The statistically insignificant decrease in extraoral time could be attributed to CBCT segmentation errors, leading to discrepancies between the 3D-printed replica and the donor tooth. These errors have been previously reported and may result from the volume-averaging effect, threshold value selection, and challenges in replicating precise landmark locations^{19,32,33}. Additionally, unlikely changes in the replica size due to sterilization or further growth of the donor tooth root between CBCT scanning and surgery could augment this discrepancy. Another integral element of our results is the potential impact on the periodontal membrane. Given the custom-made replica's precise fit, "try-ins" were not needed. Established studies agree that a shorter extra-alveolar time of the donor tooth minimizes damage to the periodontal membrane, leading to successful transplantation reported in previous case reports.³⁴ While there was no significant difference in extra-alveolar time between the groups, the replica group exhibited a shorter overall surgical time, highlighting the potential advantages and considerations of integrating CBCT, software, and 3D printing in autotransplantation for enhanced surgical efficiency.

Previous results on autotransplants using 3D-printed replicas indicated improvements during treatment, such as reduced tooth tissue damage, mostly in the remaining root follicle, and reduced extra-alveolar time of the tooth, which was 30s.³⁵ In our study, the extraoral time averaged approximately 1 min for both groups, which

was markedly less than that reported in previous literature. While some studies have noted brief extraoral times of less than 1 min with replica use,^{17,19,36} they have also reported longer durations of 3–7 min without replicas. The procedure times in these studies were substantially different based on the use of a replica, ranging from 30 to 45 min with a replica to 40–90 min without. Notably, these studies predominantly focused on premolar autotransplantation. When molars were transplanted, the extraoral time extended to an average of 7.5 min or more, particularly when extraoral endodontic procedures were involved.^{18,37} Lee et al.¹⁸ reported similar results. Both extraoral time and number of fitting attempts have been identified as prognostic indicators of transplantation success.³⁸ In contrast to the findings of EzEldeen et al.,¹⁷ our study did not find any significant differences in the number of fitting attempts per group. All autotransplantations in our study were conducted by a single surgeon who had completed more than 20 autotransplantations prior to the study. This level of expertise could account for the lack of observed differences in the number of fitting attempts between the two groups. Thus, while previous studies have emphasized various aspects of the utility of replicas, from optimizing recipient socket sizes¹⁸ to preserving both the periodontal membrane and pulp vitality,¹⁹ our results suggest that the benefit of using 3D replicas may be more pronounced for surgeons with less experience in autotransplantation procedures.³⁹

In the preliminary follow-up results averaging 1 year post-surgery, 19 out of 22 teeth in the control group and 22 out of 23 teeth in the 3D replica group met all the success criteria (no clinical/radiological signs of infection/resorption, tooth present). These preliminary data suggest that the surgical time alone does not significantly affect the success of the procedure. In future reports,

we will offer a detailed analysis of the clinical variables from this 1-year follow-up. Other factors, such as donor tooth management and surgical trauma, are equally important and warrant further investigation, possibly through the objective measurement of inflammatory markers.

This study has several limitations. All operations were performed by a single surgeon, which could have resulted in enhanced proficiency over time and possibly better outcomes. Utilizing only one operator was a methodological decision to minimize inter-operator variability, allowing for a more controlled assessment of the impact on surgical time. The intention was to isolate other variables that might affect surgical time, such as the complexity of the case or the type of procedure, to gauge the effect of the specific techniques or technologies under study more accurately. However, this limits the generalizability of our findings because each operator's unique skill set and surgical approach may influence the results, considering that in surgical trials, even standardized procedures can differ by surgeon, especially when incorporating new technologies.⁴⁰ A surgeon's learning curve also introduces variability into the outcomes.⁴⁰ Therefore, future research should involve multiple operators to broaden the understanding of the utility of 3D-printed replicas in dental autotransplantation procedures. Additionally, the development and refinement of the 3D replicas followed a learning curve. The quality of the replicas improved as our biomedical engineer accrued more experience and perceived the efficacy of the replicas. Another constraint was the lack of random patient assignment owing to operational factors linked to refining the 3D protocol. This sequential patient categorization influenced the surgeon's evolving expertise and improved the quality of the 3D replicas over time. This underscores the importance of longitudinal assessment to accurately capture the cumulative benefits and limitations of autotransplantation using 3D replicas. Despite these limitations, the integration of CBCT, software, and 3D printing into autotransplantation is a promising approach. As per the protocol of the current study, we will perform long-term evaluations of patients to document prospective clinical outcomes, including overall survival and rates of infection or alveolitis. Our findings demonstrated that the use of 3D-printed replicas in the study group significantly reduced the total surgery time. While we only measured surgical time reduction, previous studies have indicated that shorter surgeries correlate with better patient outcomes and fewer complications,⁴¹ whereas longer durations are linked to higher risks, including infections.^{42,43}

AUTHOR CONTRIBUTIONS

Miks Lejnicks: Conceptualization, investigation, methodology, project administration, validation, visualization, and review & editing. **Ilze Akota:** Conceptualization, review & editing. **Gundega Jākobsone:** Review & editing. **Laura Neimane:** Funding acquisition and review & editing. **Oskars Radzins:** Investigation, resources, software, and review & editing. **Sergio E. Uribe:** Data curation, formal analysis, investigation, methodology, visualization, validation, writing — original draft and writing — review & editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data and analysis scripts supporting this research's results are available in the RSU Institutional Dataverse <https://doi.org/10.48510/FK2/ERNRX>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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